

Marking Sheet

One sheet per student · ÆIII Mathematics — Paper Diagnostic

HOW TO MARK.

The paper cannot adapt, so you do. Work through each strand from the bottom tier up.

A STUDENT CLEARS A TIER BY GETTING MORE THAN HALF OF ITS ITEMS RIGHT.

Stop climbing at the first tier they do not clear. The strand result is the highest tier they cleared — **0** means they did not clear tier 1, and that is a real result, not a failure to record.

A tier only counts if the tier below it was cleared. A student who misses tier 1 but happens to get a tier-3 item right has guessed; do not promote them on it.

STUDENT NAME		DATE	
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Algebra

TIER	QUESTION NUMBERS — TICK EACH ONE CORRECT	CORRECT	CLEARs AT	CLEARED?
1	1 ☐ 2 ☐ 3 ☐ 4 ☐		3 of 4	
2	5 ☐ 6 ☐ 7 ☐		2 of 3	
3	8 ☐ 9 ☐ 10 ☐		2 of 3	

Algebra result: highest tier cleared = 0 1 2 3 (circle one)

Geometry & Trigonometry

TIER	QUESTION NUMBERS — TICK EACH ONE CORRECT	CORRECT	CLEARs AT	CLEARED?
1	11 ☐ 12 ☐ 13 ☐		2 of 3	
2	14 ☐ 15 ☐ 16 ☐		2 of 3	
3	17 ☐ 18 ☐		2 of 2	

Geometry & Trigonometry result: highest tier cleared = 0 1 2 3 (circle one)

Analysis

TIER	QUESTION NUMBERS — TICK EACH ONE CORRECT	CORRECT	CLEARs AT	CLEARED?
1	19 ☐ 20 ☐		2 of 2	
2	21 ☐ 22 ☐		2 of 2	
3	23 ☐		1 of 1	

Analysis result: highest tier cleared = 0 1 2 3 (circle one)

Probability & Statistics

TIER	QUESTION NUMBERS — TICK EACH ONE CORRECT	CORRECT	CLEAR AT	CLEARED?
1	24 ☐ 25 ☐		2 of 2	
2	26 ☐ 27 ☐		2 of 2	
3	28 ☐		1 of 1	

Probability & Statistics result: highest tier cleared = 0 1 2 3 (circle one)

Combinatorics

TIER	QUESTION NUMBERS — TICK EACH ONE CORRECT	CORRECT	CLEAR AT	CLEARED?
1	29 ☐		1 of 1	
2	30 ☐		1 of 1	

Combinatorics result: highest tier cleared = 0 1 2 3 (circle one)

Reading the result

TIER	WHAT IT MEANS AND WHERE THE STUDENT STARTS
0	The foundations of this strand are not in place. Start at the beginning of the strand — not at revision, at teaching.
1	Routine one-step work is secure. Start where two ideas have to be chained together.
2	Standard exam work is secure. Start on the harder material — the multi-step and unfamiliar-setup questions.
3	Secure across the strand at diagnostic level. Move to past-paper practice and use class time on the other strands.

The five strand results are independent and usually differ. A student at tier 3 in algebra and tier 0 in geometry is common and is exactly the situation this paper exists to find; teach to the lower number, not to an average of the five.

Thin tiers are fragile — treat them as provisional. Items are allocated by how much of the exam each strand is worth, so Analysis, Probability & Statistics and Combinatorics have tiers of only one or two items. On a two-item tier "more than half" means both, so a single careless slip costs a whole tier. Where a strand result rests on one or two items and it disagrees with what you see from the student in class, trust the class. Algebra and Geometry & Trigonometry carry three or four items a tier and are the two results to lean on.

What this paper does not tell you. Thirty items place a student per STRAND. They do not give a per-skill profile — for that a student needs the adaptive test, which asks a different next question depending on what they just did. Do not read a single item as evidence about a single skill.